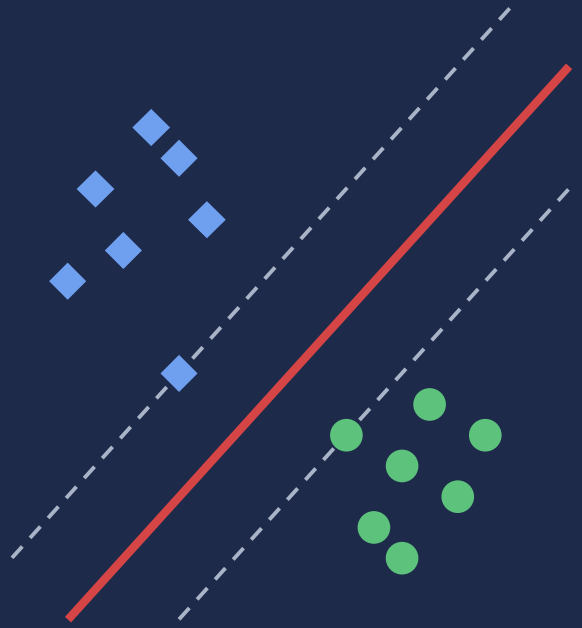
Support Vector Machine (SVM)

Hyperplane, margin, support vectors, linear and non-linear SVM, and the kernel trick

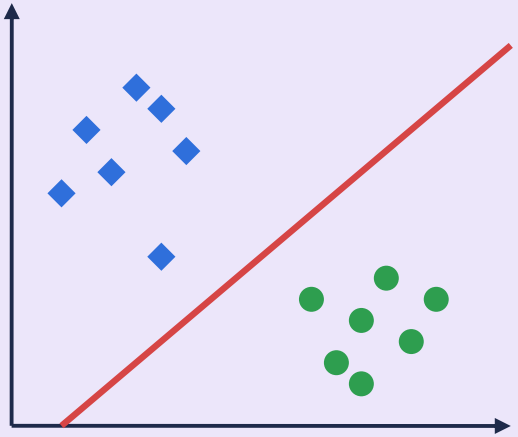


What Is SVM?

- Support Vector Machine (SVM) is one of the most popular Supervised Learning algorithms, used for Classification as well as Regression problems.
- However, it is primarily used for Classification problems in Machine Learning.
- The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n-dimensional space into classes, so that we can easily put a new data point in the correct category in the future.
- This best decision boundary is called a hyperplane.

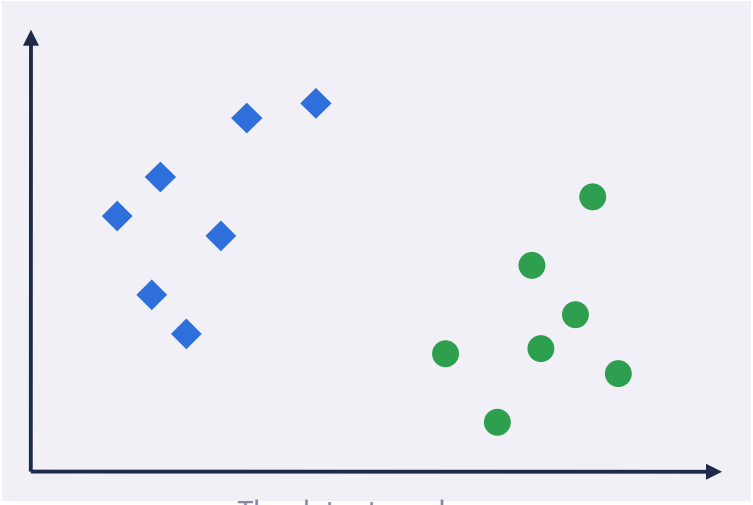
In short

SVM finds the hyperplane that best separates the classes with the maximum margin.

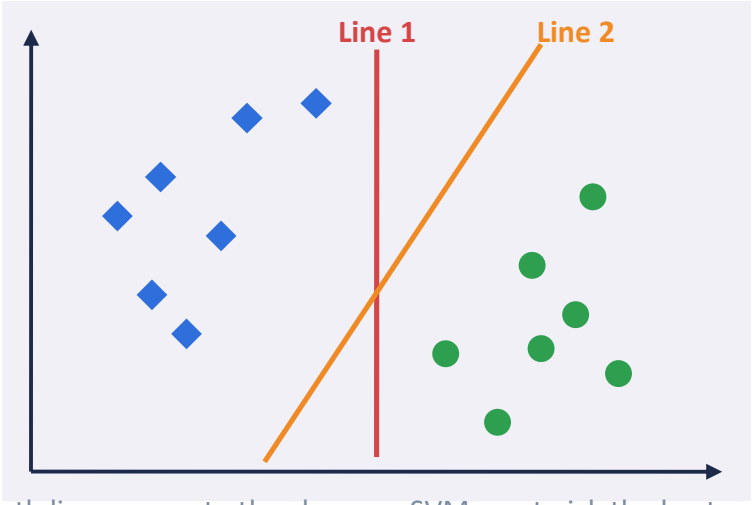


Many Possible Boundaries – Which Is Best?

- There can be multiple lines/decision boundaries to segregate the classes in n-dimensional space, but we need to find the best decision boundary that helps to classify the data points.
- This best boundary is known as the hyperplane of SVM.



The data: two classes



Both lines separate the classes – SVM must pick the best one

Key Concepts

Hyperplane

The decision boundary that separates the classes.

Support Vectors

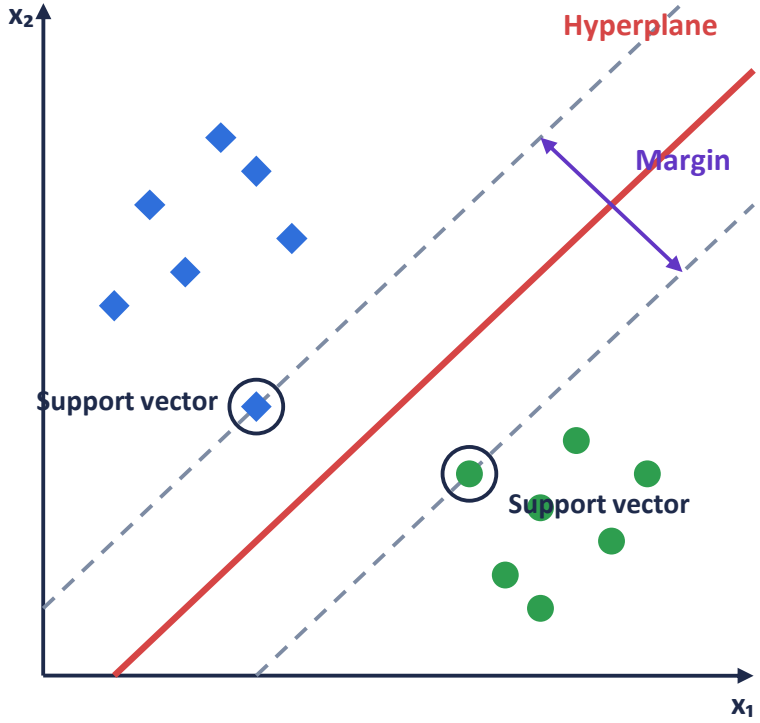
The data points nearest to the boundary.

Margin

The distance from the boundary to the support vectors.

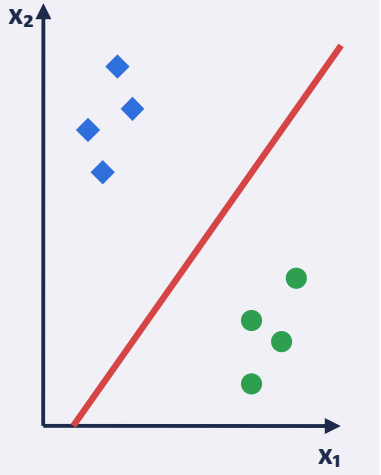
Goal

Maximise the margin.

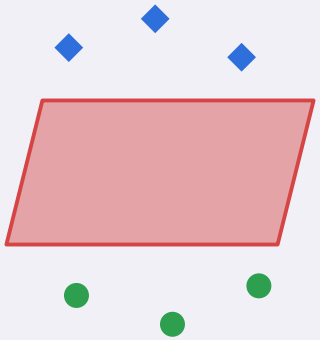


Hyperplane

- The dimensions of the hyperplane depend on the features present in the dataset: if there are 2 features, the hyperplane will be a straight line.
- If there are 3 features, the hyperplane will be a 2-dimensional plane.
- We always create a hyperplane that has a maximum margin, which means the maximum distance between the data points.



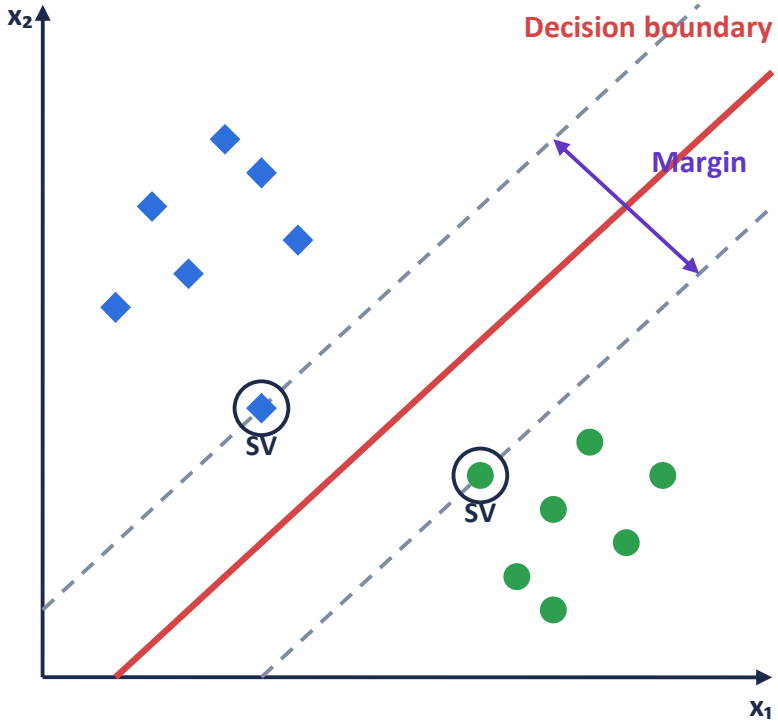
2 features
hyperplane = a line



3 features
hyperplane = a 2-D plane

In n dimensions, the hyperplane has n − 1 dimensions.

Support Vectors and the Optimal Hyperplane



- The data points or vectors closest to the hyperplane, which affect its position, are termed Support Vectors.
- Since these vectors support the hyperplane, they are called Support Vectors.

Optimal hyperplane

- The hyperplane that maximises the margin between the two classes.
- Dashed lines = margin boundaries.
- Points on the dashed lines = support vectors.
- The distance between support vectors and hyperplane should be as large as possible.

How SVM Works: Step by Step

1 **Plot the data**
Each example is a point in n-dimensional space (one axis per feature).

2 **Draw candidate boundaries**
Many lines/planes may separate the two classes.

3 **Find the support vectors**
For each boundary, find the points of each class closest to it.

4 **Measure the margin**
The gap between the boundary and the support vectors on both sides.

5 **Pick the maximum margin**
Choose the hyperplane with the largest margin – the optimal hyperplane.

6 Outliers → Soft margin (allow some errors)
Non-linear data → Kernel trick (transform to higher dimension)

7 **Classify new points**
A new point gets the class of the side of the hyperplane it falls on.

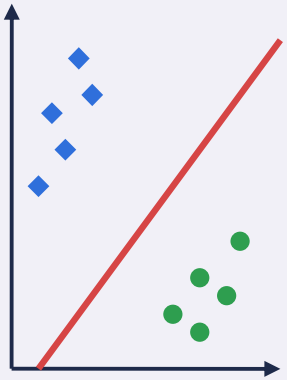
Rule: $w \cdot x + b \geq 0 \rightarrow$ class 1; $w \cdot x + b < 0 \rightarrow$ class 2

Types of SVM

SVM can be of two types:

1. Linear SVM

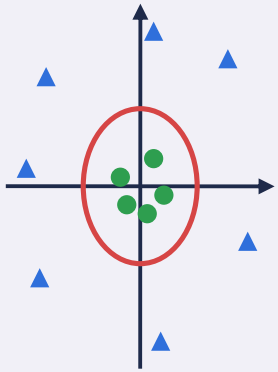
Used for linearly separable data: if a dataset can be classified into two classes using a single straight line, the data is termed linearly separable, and the classifier used is called a Linear SVM classifier.



One straight line works

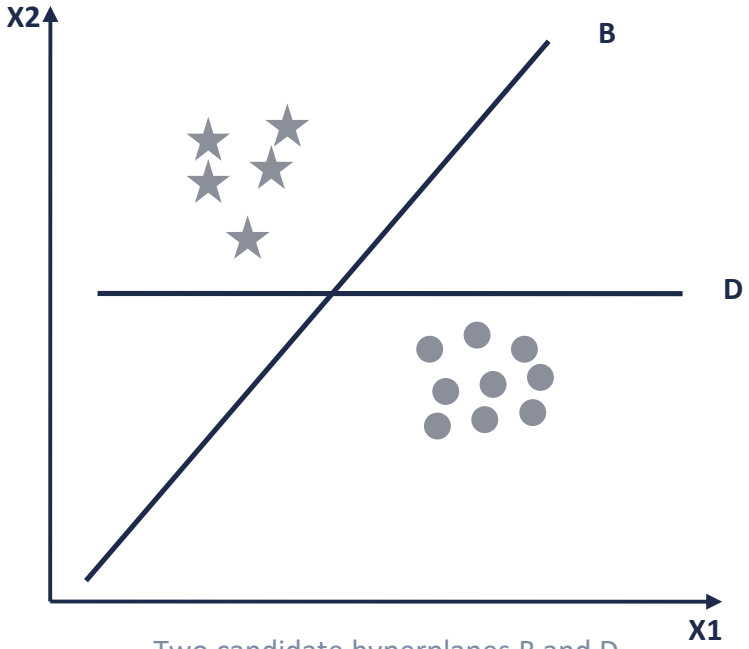
2. Non-linear SVM

Used for non-linearly separated data: if a dataset cannot be classified using a straight line, the data is termed non-linear data, and the classifier used is called a Non-linear SVM classifier.

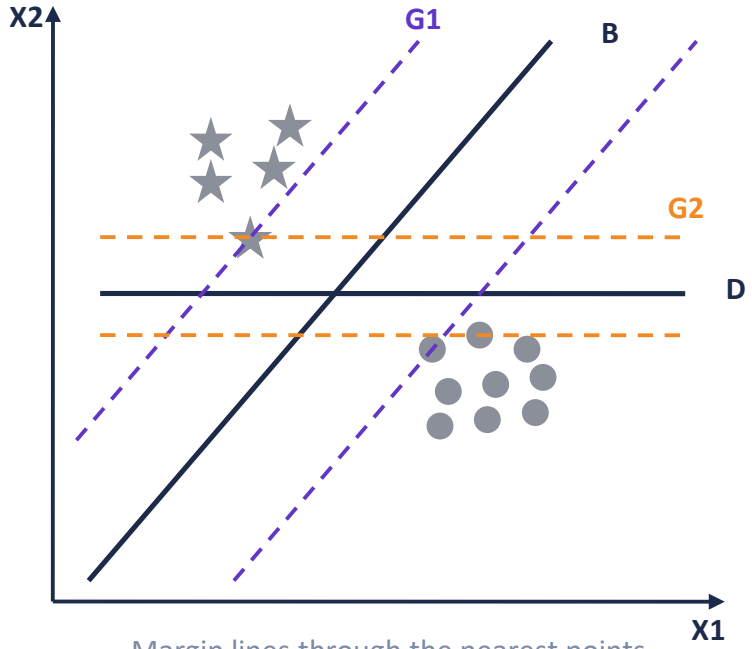


No straight line works

Linear SVM: Choosing the Best Hyperplane



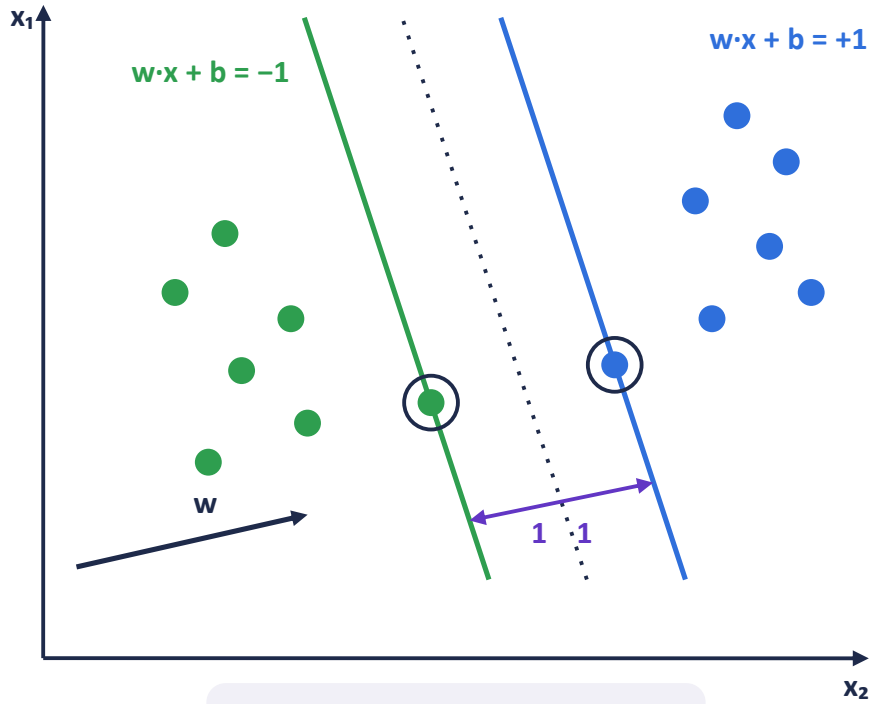
Two candidate hyperplanes B and D



Margin lines through the nearest points

Margin of B > margin of D → **B is the better hyperplane.**

The Margin in Equations



w·x + b = 0 (hyperplane)

SVM only allows hyperplanes that satisfy $y_i(w \cdot x_i + b) \geq 1$ for all points.This just says positive points must have $w \cdot x + b \geq +1$ and negative points must have $w \cdot x + b \leq -1$.

SVM optimisation problem

$$\max \frac{2}{\|\vec{w}\|}$$

$$\text{s.t. } (\vec{w} \cdot \vec{x} + b) \geq 1, \quad \forall x \text{ of class 1}$$

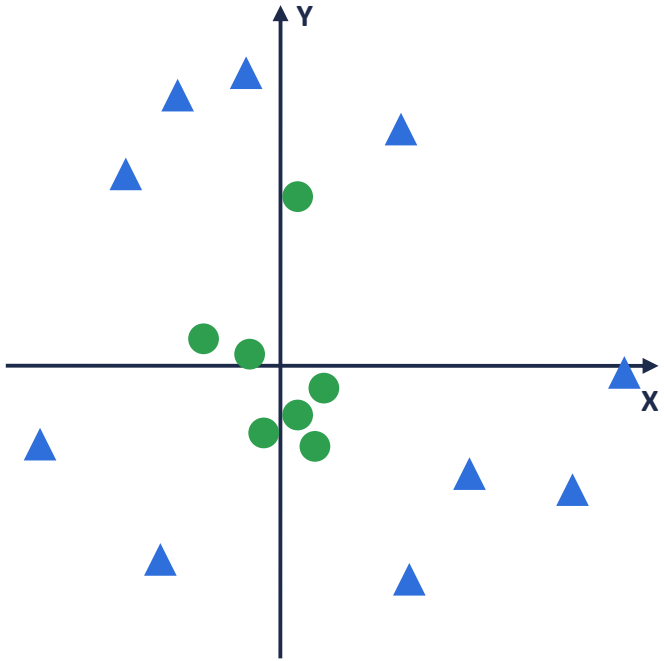
$$(\vec{w} \cdot \vec{x} + b) \leq -1, \quad \forall x \text{ of class 2}$$

- Hyperplane: $w \cdot x + b = 0$; margin lines: $w \cdot x + b = +1$ and -1.
- Each side of the margin is $1/\|w\|$, so the total margin = $2/\|w\|$.
- Maximising $2/\|w\|$ (minimising $\|w\|$) gives the widest margin while every point stays on its correct side.

Non-Linear SVM

- If data is linearly arranged, we can separate it using a straight line, but for non-linear data we cannot draw a single straight line.
- So to separate these data points, we need to add one more dimension.
- For linear data we used two dimensions x and y, so for non-linear data we will add a third dimension z.
- It can be calculated as:

$$z = x^2 + y^2$$



Green points sit in the middle, surrounded by blue – no straight line can separate them

Kernel Trick?

Definition

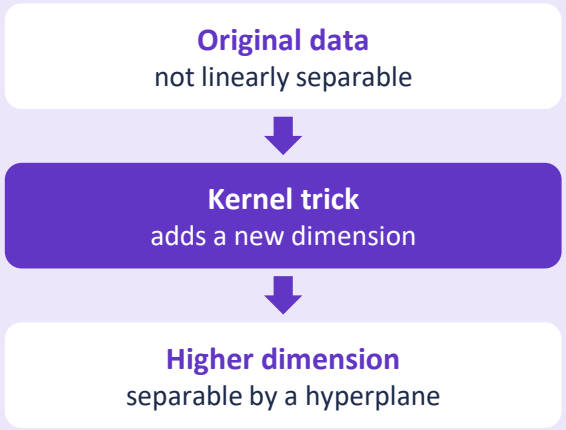
The kernel trick is a shortcut that lets SVM separate data that no straight line can split. It lifts the data into a higher dimension, where a flat boundary can separate the classes.

Purpose

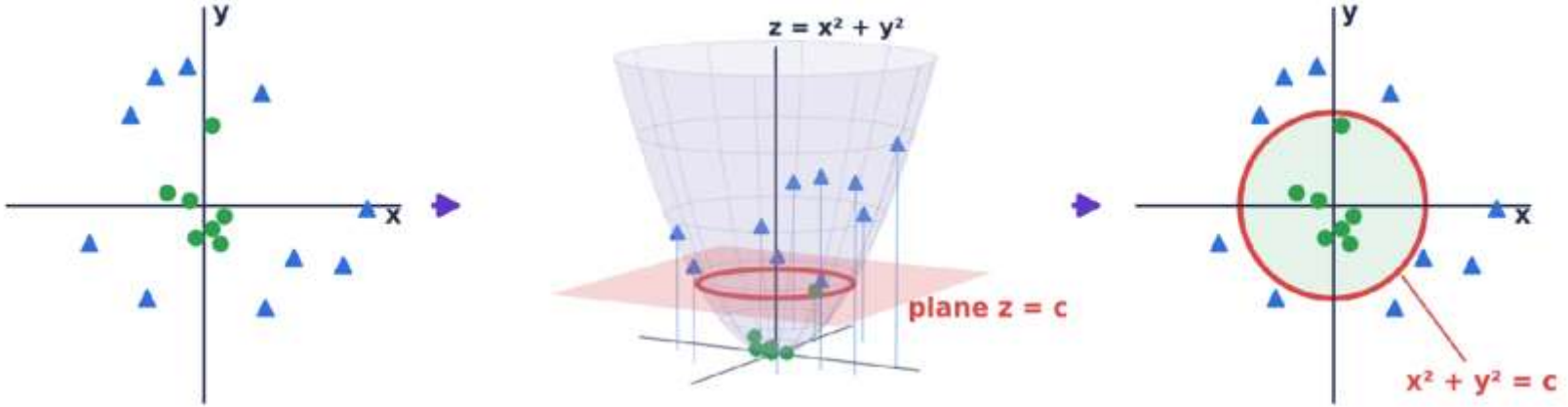
- **Handle non-linear data:** it turns a curved separation problem into a straight-line one.
- **Save time and memory:** SVM does not need to build and store the extra features.
- **Keep SVM simple:** SVM still finds the maximum-margin hyperplane, just in the new space.

In short

Lift the data so a straight boundary can separate it, without the heavy calculation.



The Kernel Trick: Adding $z = x^2 + y^2$



1 Original (x, y)

No straight line separates the classes

2 Map $\phi(x, y) = (x, y, x^2 + y^2)$

Points rise onto a bowl; a flat plane separates

3 Back in (x, y)

The plane z = c becomes a circle

What happens

Green points lie near the origin, so $z = x^2 + y^2$ is small; blue points lie far out, so z is large. A linear SVM in (x, y, z) finds the plane z = c, which is the circle $x^2 + y^2 = c$ in the original data.

Why it is a "trick"

SVM only needs dot products between points. A kernel returns $\phi(a)\cdot\phi(b)$ directly, here $K(a, b) = a\cdot b + \|a\|^2\|b\|^2$, so the new features are never built. Common kernels: polynomial, RBF.

Advantages and Limitations

Advantages

- Works well in high-dimensional spaces (many features), e.g. text and images
- Effective even when features outnumber the samples
- Memory efficient: only the support vectors define the hyperplane
- Kernel trick handles non-linear boundaries
- Maximum margin gives good generalisation; soft margin tolerates outliers

Limitations

- Slow to train on very large datasets
- Struggles when classes overlap heavily or data is noisy
- Needs careful choice of kernel and its parameters (C, γ)
- Needs feature scaling (e.g. salary vs age)
- No direct probability output, and harder to interpret than a decision tree